Home | app.py | Retail Sales Dashboard

localhost:8501

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Deploy

Dashboard Filters

Month
All

Region
All

Sales Channel
All

Product Category
All

Retail Sales Dashboard

Interactive analysis of fulfilled retail orders (completed and shipped)

Key Performance Indicators

Gross Sales	Fulfilled Orders	Average Order Value	Top Category
PHP 5,368,956.93	11,439	PHP 434.35	Toys

Net Sales after Order Discounts: PHP 4,968,543.75

Customer Count: 22,571 Repeat vs. 4,506 New

Sales Trend

Monthly Gross Sales



Key Performance Indicators

- Gross Sales: PHP 5,368,956.93
- Fulfilled Orders: 11,439
- Average Order Value: PHP 434.35
- Top Category: Toys
- Net Sales after Order Discounts: PHP 4,968,543.75
- Customer Count: 22,571 Repeat vs. 4,506 New

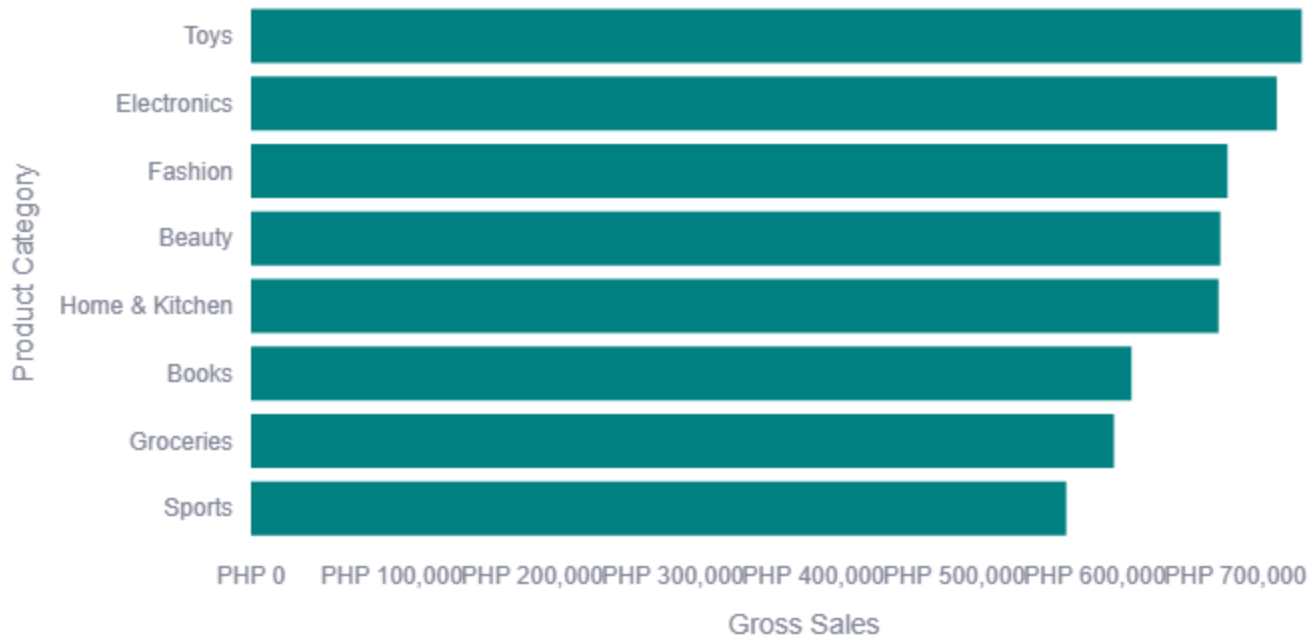
🗓️ Monthly Gross Sales



Insights and Recommendations

- Insight 1: monthly gross sales fluctuated significantly from January 2024 to January 2026, peaking at approximately PHP 239,000 in August 2025 before declining to around PHP 200,000 by January 2026, which suggests inconsistent sales performance and a weakening trend toward the end of the period.
- Recommendation 1: analyze the factors behind peak-sales months and replicate successful strategies such as effective promotions or seasonal campaigns – while investigating the late-2025 decline; moreover, implement targeted marketing and sales initiatives to stabilize monthly revenue and sustain growth.

Gross Sales by Product Category



Insights and Recommendations

- Insight 2: Toys generate the highest gross sales at approximately PHP 740,000, followed by Electronics at PHP 720,000; Sports has the lowest sales at around PHP 570,000, indicating differences in customer demand across product categories.
- Recommendation 2: prioritize inventory and marketing for high-performing categories like Toys and Electronics to sustain revenue – while reviewing the performance of Sports and other lower-selling categories to identify opportunities for improvement through targeted promotions and product assortment adjustments.

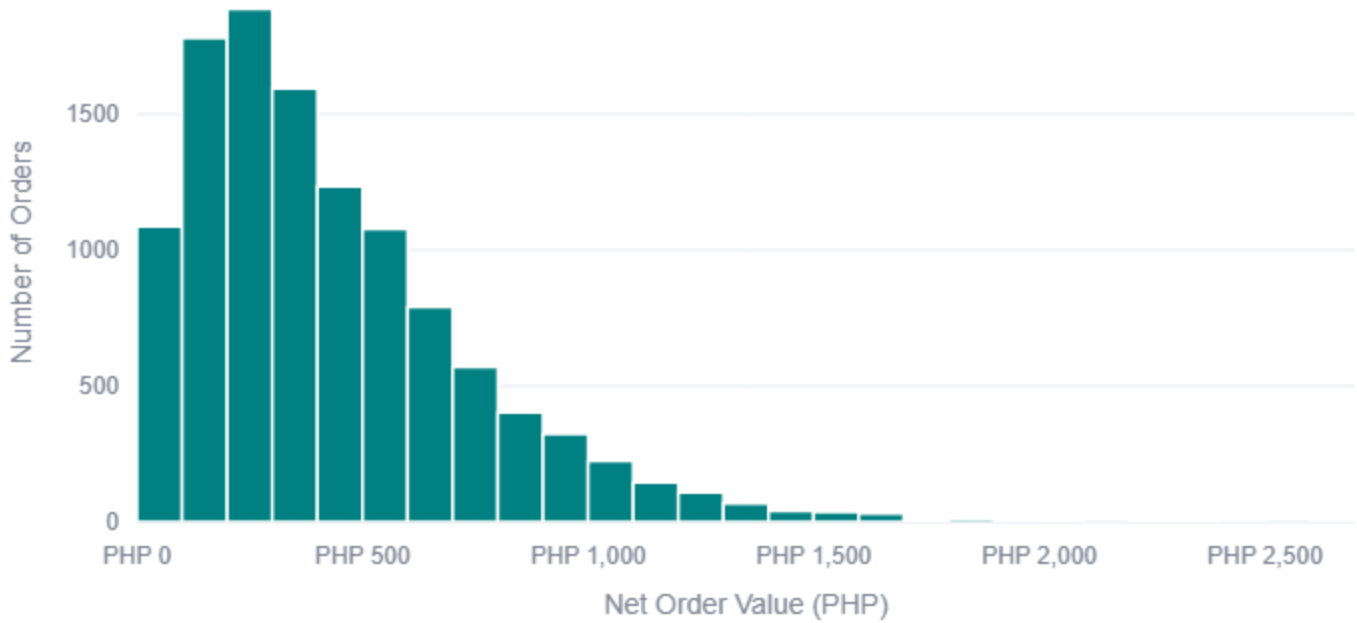
📌 Gross Sales by Region



Insights and Recommendations

- Insight 3: Central Luzon generates the highest gross sales at approximately PHP 950,000, followed by Central Visayas at PHP 925,000; CALABARZON records the lowest sales at around PHP 830,000, indicating opportunities to improve regional performance.
- Recommendation 3: maintain strong sales strategies in Central Luzon and Central Visayas – while investigating the factors behind CALABARZON's lower sales; moreover, implement targeted regional promotions and strengthen distribution efforts to increase revenue and reduce performance gaps across regions.

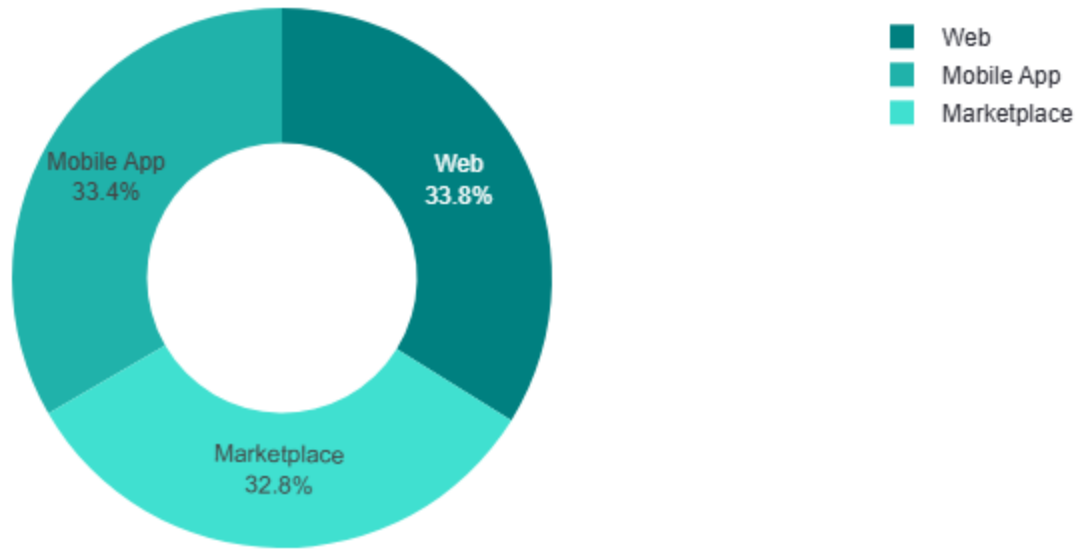
🔔 Order Value Distribution



Insights and Recommendations

- Insight 4: order values are heavily concentrated below PHP 500, with the highest frequency around PHP 200–300; the distribution is right-skewed, meaning high-value orders are relatively rare.
- Recommendation 4: focus on increasing average order value through targeted upselling, product bundles, and free-shipping incentives for orders above PHP 500 – while maintaining affordable options that appeal to the majority of customers.

🛒 Gross Sales by Sales Channel



Insights and Recommendations

- Insight 5: the Web channel generates the highest gross sales, whereas the Marketplace has the lowest; however, sales across all three channels are relatively similar at approximately PHP 1.75–1.80 million.
- Recommendation 5: maintain strong performance across all channels – while analyzing the Web channel's successful strategies and applying them to the Marketplace to help close the sales gap and increase overall revenue.

Monthly Return Count



Insights and Recommendations

- Insight 6: monthly returns fluctuate significantly, ranging from approximately 67 to 119 — with notable peaks in June 2024, November 2024, and July 2025; despite these spikes, returns declined toward the end of 2025 before recovering in January 2026.
- Recommendation 6: investigate the factors driving peak-return months and the sharp declines, then use these insights to develop targeted strategies to stabilize monthly returns and replicate successful performance patterns.

Top 10 Products by Gross Sales



Insights and Recommendations

- Insight 7: KidJoy Toys 607 is the top-selling product by gross sales, significantly outperforming the other products, whereas BookNest Groceries 219 ranks lowest among the top 10; this indicates stronger sales performance in certain product categories, particularly toys.
- Recommendation 7: prioritize inventory and marketing for high-performing products like KidJoy Toys 607 to sustain sales momentum – while reviewing the pricing, promotions, and customer demand for lower-performing products to identify opportunities for improvement.

Summary and Conclusions

- Toys emerged as the leading product category, with KidJoy Toys 607 achieving the highest product sales, whereas Sports recorded the lowest category sales.
- Geographically, Central Luzon performed best, whereas CALABARZON offers opportunities for improvement.
- The Web channel generated the highest gross sales, although sales across all channels remained relatively balanced.

Summary and Conclusions

- Most orders were valued below PHP 500, suggesting that the business primarily serves budget-conscious customers.
- Monthly sales fluctuated considerably, reaching a peak in August 2025 before declining toward January 2026, whereas product returns also varied throughout the period.
- Customer data shows a strong repeat-customer base, indicating potential customer loyalty and opportunities to attract new buyers.

Summary and Conclusions

- Overall, the business demonstrates solid sales performance and a strong repeat-customer base; however, declining recent sales, fluctuating returns, and uneven category and regional performance highlight the need for targeted marketing, improved customer acquisition, effective return management, and strategies to increase average order value and sustain long-term growth.

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PHP 0 PHP 5,000 PHP 10,000 PHP 15,000 PHP 20,000 PHP 25,000

Gross Sales

Filtered Transaction Details

order_id	order_date	product_name	category_clean	quantity	unit_price_clean	region_name	city	sales_channel	discount_pct	gross_sales	net_sales
ORD0001460	2026-01-31 23:36:00	FitPro Groceries 390	Groceries	1	117.74	Central Visayas	Cebu City	Web	0	117.74	117.74
ORD0001460	2026-01-31 23:36:00	Acme Home 695	Home & Kitchen	1	76.62	Central Visayas	Cebu City	Web	0	76.62	76.62
ORD0001460	2026-01-31 23:36:00	FreshCart Beauty 175	Beauty	1	97.69	Central Visayas	Cebu City	Web	0	97.69	97.69
ORD0001460	2026-01-31 23:36:00	BookNest Toys 579	Toys	1	134.22	Central Visayas	Cebu City	Web	0	134.22	134.22
ORD0005059	2026-01-31 22:08:00	SunPeak Groceries 527	Groceries	1	61.87	Caraga	Surigao City	Mobile App	0.075	61.87	57.2298
ORD0005059	2026-01-31 22:08:00	Acme Sports 177	Sports	2	23.64	Caraga	Surigao City	Mobile App	0.075	47.28	43.734
ORD0005059	2026-01-31 22:08:00	FreshCart Home 376	Home & Kitchen	2	192.58	Caraga	Surigao City	Mobile App	0.075	385.16	356.273
ORD0005059	2026-01-31 22:08:00	HomeEase Books 258	Books	1	85.54	Caraga	Surigao City	Mobile App	0.075	85.54	79.1245
ORD0014802	2026-01-31 21:58:00	FreshCart electronics 129	Electronics	1	154.7	CALABARZON	Cavite City	Marketplace	0.155	154.7	130.7215
ORD0014802	2026-01-31 21:58:00	Acme Toys 958	Toys	2	127.73	CALABARZON	Cavite City	Marketplace	0.155	255.46	215.8637

Insights and Recommendations

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> Insights and Recommendations

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This dashboard uses the original retail CSV files.

The application performs basic cleaning before analysis:

- removes exact duplicate records
- standardizes product categories
- converts unit prices to numeric values
- removes zero or negative quantities
- parses mixed order-date formats
- fills missing discounts with zero
- validates customer, order, and product references
- keeps completed and shipped orders for sales analysis

Gross Sales = Quantity × Unit Price

Net Sales = Gross Sales × (1 − Discount Percentage)

The average order value is calculated at the order level, not at the order-item level.

Filtered item rows: 27,077

Filtered unique orders: 11,439

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Capstone Project

- app.py
- app.txt
- Day01_Python.ipynb
- Day02_Python.ipynb
- Day03_Python.ipynb
- Day04_Python.ipynb
- Day05_Python.ipynb
- Days01–06_Python.ipynb

Capstone Project

- _order_summary_day5.csv
- _sales_analysis_day5.csv
- customers.csv
- customers_clean.csv
- order_items.csv
- orders.csv
- orders_clean.csv
- payments.csv
- products.csv
- products_clean.csv
- returns.csv
- sales_analysis.csv
- shipments.csv

Capstone Project

- -gross-sales-by-product-category.png
- gross-sales-by-region.png
- gross-sales-by-sales-channel.png
- monthly-gross-sales.png
- monthly-return-count.png
- order-value-distribution.png
- top-10-products-by-gross-sales.png

Capstone Project

GitHub Repository Link:

https://github.com/allyster-endozo/DSA_Python